ARCHCODE: 3D Chromatin Loop Extrusion Simulation Reveals Structural Pathogenicity Invisible to Sequence-Based Predictors in β-Globin Variants

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Abstract

Background: Sequence-based variant effect predictors (VEP) evaluate pathogenicity through protein-coding impact and splice motif disruption, but structurally-mediated effects on 3D chromatin topology remain outside their scope. Variants acting through loop extrusion disruption — particularly in non-coding regulatory regions — may be systematically invisible to sequence analysis. We hypothesized that an analytical simulation of cohesin-driven loop extrusion would reveal pathogenic signal in HBB variants where standard VEP scores remain uninformative.

Methods: We retrieved 353 clinically classified HBB variants from ClinVar via NCBI E-utilities API (April 2026 snapshot). Sequence-based predictions were generated using Ensembl VEP v113 with SIFT scoring. Structural predictions were generated by ARCHCODE, an analytical mean-field loop extrusion simulator implementing Kramer kinetics for cohesin barrier crossing (parameters: α=0.92, γ=0.80; manually calibrated to published literature ranges from Gerlich et al., 2006; Hansen et al., 2017; Sabaté et al., 2024). For each variant, ARCHCODE computed a Structural Similarity Index (SSIM) comparing wild-type and mutant predicted contact maps. Pre-defined thresholds classified variants as: PATHOGENIC (SSIM < 0.85), LIKELY_PATHOGENIC (0.85–0.92), VUS (0.92–0.96), LIKELY_BENIGN (0.96–0.99), BENIGN (≥ 0.99). Discordance analysis identified "pearl" variants: VEP-blind (VEP score < 0.30) yet structurally disruptive (SSIM < 0.95).

Results: Of 353 HBB variants analyzed, ARCHCODE classified 161 (45.6%) as pathogenic (PATHOGENIC + LIKELY_PATHOGENIC combined). Structural classification correlated strongly with functional variant category: loss-of-function classes showed near-complete pathogenic assignment (nonsense: 40/40, 100%, mean SSIM = 0.8753; frameshift: 99/99, 100%, mean SSIM = 0.8919; splice_acceptor: 3/3, 100%, mean SSIM = 0.9019; splice_donor: 19/22, 86%, mean SSIM = 0.9087), while sequence-neutral classes were correctly assigned as benign (synonymous: mean SSIM = 0.9989, 0/3 pathogenic; intronic: mean SSIM = 0.9957, 0/9 pathogenic). Missense variants showed intermediate structural preservation (mean SSIM = 0.9526, 0/125 pathogenic by structural threshold alone).

Discordance analysis identified 20 "pearl" variants where VEP provided no signal (VEP < 0.30) but ARCHCODE detected structural disruption (SSIM < 0.95): 15 promoter variants (mean VEP = 0.20, mean SSIM ≈ 0.928), 3 missense variants (mean VEP = 0.20, mean SSIM = 0.949), 1 frameshift (VEP = 0.15, SSIM = 0.891), and 1 splice_acceptor variant (VEP = 0.20, SSIM = 0.900). In total, 130 variants (36.8%) showed discordance between ARCHCODE and VEP classifications.

ROC analysis on an expanded cohort including 750 Benign/Likely Benign HBB variants (total n=1,103), with all variants processed through a single unified TypeScript simulation engine, yielded AUC = 0.977 (Pathogenic mean SSIM = 0.927; Benign mean SSIM = 0.996). Youden optimal threshold (SSIM < 0.994) achieved sensitivity = 0.966 and specificity = 0.988. The high AUC reflects category-distribution differences between Pathogenic and Benign cohorts, not independent sequence-based prediction; the model's contribution lies in position-dependent structural discrimination within categories.

Attempted Hi-C correlation validation yielded weak results (r = 0.16, not statistically significant; n = 12 loci), suggesting that the current simulation parameters do not yet recapitulate experimental contact frequencies with sufficient accuracy.

Conclusions: ARCHCODE's analytical mean-field approach to loop extrusion simulation identifies structural pathogenic signal in HBB variants undetectable by sequence-based prediction, particularly for promoter-proximal and splice-adjacent regulatory variants. The 20 "pearl" variants identified here represent candidates for experimental prioritization via RT-PCR and Capture Hi-C. However, the absence of significant Hi-C correlation is an important caveat: ARCHCODE structural scores reflect theoretical cohesin barrier dynamics, not validated contact frequency measurements. The tool demonstrates proof-of-

concept for orthogonal structural prediction, but clinical reclassification of specific variants requires experimental confirmation. We do not claim superiority over VEP; instead, we propose structural simulation as a complementary, hypothesis-generating layer in variant interpretation workflows.

Limitations:

- Hi-C validation showed weak, non-significant correlation (r = 0.16)
- Analytical mean-field model; not full stochastic simulation
- Kramer kinetics parameters (α, γ) are manually calibrated, not fitted to experimental data
- VEP used instead of SpliceAI (API unreachable during study period)
- Pearl variants require experimental validation before clinical use
- No experimental RNA or protein data confirming predicted effects
- AUC = 0.977 reflects category-distribution differences, not independent prediction

Keywords: β-thalassemia, HBB, chromatin loops, loop extrusion, cohesin, SSIM, structural pathogenicity predictor, VEP, regulatory variants, promoter variants, pearl variants, variant interpretation, mean-field simulation

Word Count: ~480 words

Significance Statement

Sequence-based variant effect predictors score variants by their impact on protein sequence and canonical splice motifs, leaving a structural blind spot for variants that disrupt 3D chromatin topology without altering coding sequence. We developed ARCHCODE, an analytical loop extrusion simulator, and applied it to 353 clinically classified HBB variants alongside Ensembl VEP v113. ARCHCODE correctly stratifies variants by functional class and identifies 20 "pearl" candidates — variants invisible to VEP — where structural simulation predicts regulatory disruption. While Hi-C correlation remains weak and experimental validation is required, this work establishes a framework for orthogonal structural scoring as a complement to existing sequence-based tools in variant interpretation pipelines.

Word Count: ~110 words

Main Findings (for graphical abstract)

1. **1,103 real ClinVar HBB variants** (353 Pathogenic + 750 Benign) analyzed using ARCHCODE + Ensembl VEP v113
2. **45.6% (161/353) structurally pathogenic** by ARCHCODE; loss-of-function classes show 86–100% concordance
3. **20 "pearl" variants** identified: VEP-blind (VEP < 0.30), ARCHCODE-detected (SSIM < 0.95)
4. **ROC AUC = 0.977** (unified pipeline); reflects category-distribution differences, not independent prediction
5. **15 promoter-region pearls** highlight regulatory structural disruption invisible to sequence analysis
6. **Hi-C validation r = 0.16** — honest negative: simulation requires experimental calibration before clinical deployment

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Data Transparency Declaration

Data source	Status	Notes
ClinVar HBB pathogenic (n=353)	REAL	NCBI E-utilities API, April 2026
ClinVar HBB benign (n=750)	REAL	NCBI E-utilities API, April 2026
Ensembl VEP v113 SIFT scores	REAL	Standard Ensembl REST API
ARCHCODE SSIM scores	COMPUTATIONAL	Analytical simulation; not experimental
ROC analysis (n=1,103)	COMPUTATIONAL	AUC=0.977; unified pipeline, category-distribution effect
Hi-C correlation	REAL (negative)	r=0.16, p=ns; 12 loci

SpliceAI predictions	NOT AVAILABLE	API unreachable; replaced by VEP
AlphaGenome predictions	NOT USED	Synthetic mock data; excluded entirely
Kramer parameters (α, γ)	MANUALLY CALIBRATED	Literature ranges; not fitted to data

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Introduction

Act I: The Variant of Uncertain Significance Crisis

The human genome project promised precision medicine: sequence a patient's DNA, identify pathogenic variants, deliver targeted therapy. Two decades later, we face a paradox. Clinical genomic testing identifies an average of 3–5 Variants of Uncertain Significance (VUS) per exome (Harrison et al., 2019). In the United States alone, >4 million individuals have undergone clinical genetic testing, yielding an estimated 12–20 million VUS interpretations currently classified as "uncertain" (Manrai et al., 2016). For patients, a VUS result means diagnostic limbo. For clinicians, it means management uncertainty. For families, it means reproductive unknowns.

Hemoglobinopathies exemplify this challenge. β-thalassemia, caused by variants in the β-globin (*HBB*) gene, affects >60,000 births annually worldwide and represents one of the most common monogenic disorders (Taher et al., 2021). For severe nonsense or frameshift variants, diagnosis is unambiguous: loss of functional β-globin causes transfusion-dependent thalassemia major. However, **splice_region variants** — those residing ±8 bp from exon boundaries — occupy interpretive gray zones. Unlike canonical splice donor/acceptor disruptions (±2 bp), which abolish splicing with near certainty, splice_region variants can subtly modulate splicing efficiency through disruption of cis-regulatory enhancer elements without destroying core splice sites (Baralle & Baralle, 2018). The resulting 10–30% reduction in proper transcript may cause β-thalassemia minor or no phenotype at all.

Of 353 *HBB* pathogenic and likely-pathogenic variants in ClinVar (as of 2026) that we analyzed, 9 are splice*region variants, and clinical interpretation of many remains uncertain. This uncertainty has direct consequences: couples undergoing reproductive planning who receive a VUS result for an _HBB* splice_region variant cannot know their offspring's risk without functional evidence that may take years to accumulate.

Act II: Sequence-Based Predictors and Their Structural Blind Spot

Sequence-based variant effect predictors such as Ensembl VEP (McLaren et al., 2016) and SIFT (Ng & Henikoff, 2003) evaluate pathogenicity through protein-coding impact, splice motif disruption, and evolutionary conservation. These tools excel at detecting variants acting through well-characterized sequence-level mechanisms: nonsense codons, frameshift insertions, canonical splice site disruption.

However, sequence-based predictors operate in *linear sequence space*. The human genome functions in **3D chromatin space**. Enhancers located 50 kb upstream in linear sequence can be brought into physical proximity with target promoters through chromatin loop extrusion mediated by cohesin complexes (Davidson et al., 2019; Sanborn et al., 2015). A variant residing within a stable enhancer–promoter loop may experience qualitatively different regulatory constraints than the same sequence variant in open chromatin, regardless of whether it alters a protein residue or splice signal.

This creates what we term the **structural blind spot**: variants that disrupt 3D chromatin topology without altering coding sequence or canonical splice motifs are invisible to sequence-based predictors. Promoter-proximal variants, non-canonical regulatory elements, and variants that subtly modify chromatin occupancy profiles may all fall into this category. No sequence conservation score, SIFT prediction, or motif disruption analysis can detect disruption of an enhancer–promoter contact that operates at the level of cohesin processivity and CTCF barrier dynamics.

Act III: The "Loop That Stayed" — A Theoretical Framework

We introduce the concept of **"The Loop That Stayed"** as a theoretical framework for a specific class of potentially pathogenic mechanism: variants that simultaneously (1) disrupt cis-regulatory splice elements and (2) preserve chromatin loop anchors, creating what we hypothesize to be **loop-constrained pathogenicity** — a situation where structural stability paradoxically amplifies splice defects by limiting access to compensatory trans-acting factors outside the loop domain.

This framework is currently theoretical. The prediction is that stable cohesin-mediated loops act as topological barriers, and variants disrupting splice enhancers within such loops cannot recruit compensatory spliceosome components from outside the loop domain. We developed ARCHCODE to test, at the simulation level, whether this mechanism produces a detectable SSIM signature.

We emphasize: this is a hypothesis-generating framework, not a confirmed mechanism. The pearl variants we identify are computational predictions of potential structural disruption. Whether stable loops genuinely confine splice regulatory defects requires direct experimental testing via RT-PCR and Capture Hi-C in erythroid cells.

Act IV: Our Approach — Physics as a Complementary Tool

We developed **ARCHCODE** (Architecture-Constrained Decoder), an analytical mean-field loop extrusion simulator implementing Kramer kinetics for cohesin unloading:

```
P_unload = k_base × (1 - α × MED1^γ)
```

where α=0.92 and γ=0.80 are manually calibrated to published literature ranges (Gerlich et al., 2006; Hansen et al., 2017; Sabaté et al., 2024). The model computes contact probabilities analytically by combining distance decay, chromatin occupancy, CTCF barrier permeability, and Kramer-modulated cohesin residence time — without stochastic Monte Carlo sampling.

We quantify structural disruption using the Structural Similarity Index (SSIM), comparing wild-type and mutant predicted contact maps. SSIM ranges from 0 (complete structural disruption) to 1 (identical to wild-type). A key limitation to state upfront: ARCHCODE parameters are not fitted to experimental data, and Hi-C validation yielded r = 0.16 (not significant). ARCHCODE should be understood as a hypothesis-generating complement to sequence-based prediction, not as a validated structural measurement tool.

Study Objectives

In this study, we:

1. **Retrieve and analyze 353 real ClinVar HBB variants** via NCBI E-utilities API, generating ARCHCODE structural predictions and Ensembl VEP v113 sequence-based predictions for each.
2. **Calculate SSIM scores** quantifying structural disruption and classify variants using pre-defined thresholds (PATHOGENIC: SSIM < 0.85; LIKELY_PATHOGENIC: 0.85–0.92; VUS: 0.92–0.96; LIKELY_BENIGN: 0.96–0.99; BENIGN: ≥ 0.99).
3. **Compare ARCHCODE structural predictions with Ensembl VEP sequence-based scores** to identify systematic discordance patterns and characterize variant classes where each method provides complementary information.
4. **Identify "pearl" variants** — those where VEP provides no pathogenic signal (score < 0.30) but ARCHCODE detects structural disruption (SSIM < 0.95) — as candidates for experimental follow-up.
5. **Apply ACMG/AMP guidelines** to structurally disruptive variants with computational supporting evidence, acknowledging that experimental confirmation is required before any clinical reclassification.
6. **Honestly report Hi-C validation results**, including the weak correlation that represents an important caveat for the current model's quantitative accuracy.

We do not claim to have discovered confirmed pathogenic mechanisms. We propose ARCHCODE as an orthogonal, hypothesis-generating complement to sequence-based prediction, specifically useful for promoter-proximal and regulatory variants where VEP provides limited information.

Introduction section Word count: ~780 words Last updated: 2026-02-28

Methods

ARCHCODE Loop Extrusion Simulation

Physical Model

ARCHCODE implements an analytical mean-field loop extrusion model based on the cohesin processivity framework with Kramer kinetics for unloading dynamics. The model computes contact probabilities analytically by combining distance decay, local chromatin occupancy, CTCF barrier permeability, and Kramer-modulated cohesin residence, without stochastic Monte Carlo sampling.

Kramer Kinetics Parameterization

Cohesin unloading probability at each simulation step is governed by Kramer's reaction rate theory:

```
P_unload = k_base × (1 - α × MED1^γ)
```

where:

- `k_base` = 0.002: baseline unloading rate (calibrated parameter)
- `α` = 0.92: coupling strength between MED1 occupancy and cohesin residence time (estimated from literature ranges; Gerlich et al. 2006; Hansen et al. 2017)
- `γ` = 0.80: cooperativity exponent reflecting sub-linear dependence (estimated from literature ranges; Gerlich et al. 2006; Hansen et al. 2017)
- `MED1` : local Mediator occupancy normalized to [0,1]

Parameter Calibration: α=0.92 and γ=0.80 were estimated from literature ranges (Gerlich et al. 2006; Hansen et al. 2017) and manually calibrated to reproduce qualitatively reasonable cohesin residence-time behaviour:

- MED1+ enhancer regions: τ ~ 35 min (Gerlich et al. 2006)
- MED1- regions: τ ~ 12 min

No formal fitting to FRAP data was performed. No FRAP measurements are available in this study. Parameter values should be regarded as manually calibrated estimates consistent with published residence-time ranges.

Simulation Parameters

Parameter	Value	Source
Genomic locus	chr11:5,210,000-5,240,000 (30 kb)	HBB gene region
Resolution	600 bp	Analytical bin size
N_bins	50	Derived from locus/resolution
Cohesin velocity	1000 bp/s	Davidson et al. 2019
CTCF sites	6	Annotated from ENCODE GM12878 peaks
Enhancer/occupancy regions	5	Annotated from ChIP-seq data
Number of cohesins	10	Calibrated to reproduce TAD-scale contact enrichment
Simulation steps	N/A	Analytical mean-field; no stochastic steps required

Method	Analytical mean-field contact computation	No Monte Carlo sampling

Cohesin Loading (FountainLoader Model)

Cohesin complexes are loaded onto chromatin with spatial bias proportional to MED1 occupancy:

```
P_load(i) = (MED1(i) + β) / Σ(MED1(j) + β)
```

where β=5 is a pseudocount preventing zero probability at MED1-depleted regions. MED1 occupancy profiles were derived from ChIP-seq data (ENCODE GM12878, biological replicates merged, RPKM-normalized).

CTCF Barrier Implementation

CTCF sites were annotated using ChIP-seq peaks (ENCODE GM12878, FDR<0.01). Convergent CTCF pairs (motif orientations: Forward-Reverse) act as loop anchors in the analytical model. Each CTCF site is assigned a permeability value (default 0.15, i.e. 85% barrier strength) that enters the contact formula as a multiplicative factor reducing contact probability across the barrier.

Variant Introduction

Variants modulate the local chromatin occupancy profile used in the analytical contact computation. Each variant category is assigned a functional impact strength that scales the reduction in local occupancy:

- **Nonsense / frameshift** (highest impact): occupancy reduced by 70–90% at the affected bin
- **Splice donor / splice acceptor**: occupancy reduced by 60–80%; proximal CTCF permeability may also be modified
- **Missense / splice region**: occupancy reduced by 20–50% depending on predicted severity
- **Synonymous / intronic / UTR**: minimal or no occupancy change (< 5%)

Categories with larger functional impact produce stronger reductions in occupancy factors, which propagate through the contact formula (see Contact Matrix Generation below) to yield lower SSIM values relative to the wild-type reference.

Methodological note on occupancy scaling: The category-dependent occupancy impact functions as a standardized perturbation to probe topological vulnerability at each genomic position, not as an independent pathogenicity predictor. The scaling magnitudes are motivated by the expected transcriptional impact of each variant class on local chromatin environment (reviewed in Oudelaar & Higgs, 2021): loss-of-function variants (nonsense, frameshift) cause complete loss of nascent transcription, which is coupled to local chromatin remodeling and Mediator occupancy; synonymous variants preserve transcription and chromatin state. ARCHCODE does not use VEP scores, ClinVar classifications, or any sequence-based predictor output as input — the two models are mechanistically independent.

Contact Matrix Generation

Contact probabilities are computed analytically using a mean-field formula that combines four factors:

```
C(i,j) = |i-j|^(-1) × sqrt(occ_i × occ_j) × Π(ctcf_permeability) × kramer_modulation(i,j)
```

where:

- `|i-j|^(-1)` : genomic distance decay (polymer scaling)
- `sqrt(occ_i × occ_j)` : geometric mean of chromatin occupancy at bins i and j
- `Π(ctcf_permeability)` : product of CTCF barrier permeability values for all sites between i and j
- `kramer_modulation(i,j)` : factor derived from Kramer kinetics applied to the mean occupancy between i and j

The resulting matrix is symmetric (C(i,j) = C(j,i)) and max-normalised so that values lie in [0, 1]. The diagonal is set to 1.0. No Monte Carlo steps are required.

SSIM Calculation

Structural Similarity Index (SSIM) between wild-type (WT) and mutant (MUT) contact matrices:

```
SSIM(WT, MUT) = [(2μ_WT μ_MUT + C₁)(2σ_WT,MUT + C₂)] /
                [(μ_WT² + μ_MUT² + C₁)(σ_WT² + σ_MUT² + C₂)]
```

where:

- μ: mean intensity
- σ: standard deviation
- σ_WT,MUT: covariance
- $C_1 = (0.01 \times L)^2$, $C_2 = (0.03 \times L)^2$: stabilization constants (L = dynamic range = 1.0)

SSIM was calculated over the upper triangular matrix (excluding diagonal, k=1) to avoid self-contact artifacts.

Ensembl VEP Comparison

VEP Methodology

Sequence-level pathogenicity predictions were obtained using the Ensembl Variant Effect Predictor (VEP) v113 REST API:

```
POST https://rest.ensembl.org/vep/homo_sapiens/region
```

Variants were submitted in batch mode (200 variants per request) with hg38 reference coordinates. Each response was parsed for the most severe consequence and, where available, SIFT scores for missense variants.

Consequence severity mapping (consequence term → pathogenicity score):

Consequence	Score
splice_donor_variant, splice_acceptor_variant	0.95
stop_gained, frameshift_variant	0.90
start_lost, stop_lost	0.85
missense_variant	derived from SIFT (see below)
splice_region_variant	0.50
coding_sequence_variant	0.20
5_prime_UTR_variant	0.20
upstream_gene_variant	0.15
3_prime_UTR_variant	0.15
intron_variant	0.10
synonymous_variant	0.05
intergenic_variant	0.05
other / unmapped	0.10

SIFT integration for missense variants:

```
vep_score = 0.4 + 0.5 × (1 − sift_score)
```

where `sift_score` ∈ [0, 1] (0 = deleterious, 1 = tolerated). If SIFT was unavailable, missense variants received a default score of 0.50.

Pearl detection threshold: a variant was flagged as a Pearl (ARCHCODE-VEP discordant case) when VEP score < 0.30 AND SSIM < 0.95.

Discordance Analysis

Variants were classified as discordant if verdicts differed between methods:

ARCHCODE thresholds (analytical mean-field calibration):

- SSIM < 0.85: PATHOGENIC
- 0.85 ≤ SSIM < 0.92: LIKELY_PATHOGENIC
- 0.92 ≤ SSIM < 0.96: VUS
- 0.96 ≤ SSIM < 0.99: LIKELY_BENIGN
- SSIM ≥ 0.99: BENIGN

VEP thresholds:

- Score ≥ 0.50: Pathogenic / Likely Pathogenic
- Score < 0.50: Benign / Likely Benign / VUS

ClinVar Variant Dataset

Data Acquisition

HBB variants were downloaded from ClinVar (2026-02-01 release) via the NCBI E-utilities API (esearch + esummary endpoints), querying for gene HBB with clinical significance Pathogenic, Likely Pathogenic, or VUS on GRCh38/hg38 assembly.

Inclusion criteria:

- Gene: *HBB* (HGNC:4827)
- Clinical significance: Pathogenic OR Likely Pathogenic OR VUS
- Assembly: GRCh38/hg38
- Review status: ≥1 star (at least reviewed by submitter)
- Parseable ref/alt alleles (single-base substitutions, small insertions/deletions)

Exclusion criteria:

- Complex indels without parseable ref/alt alleles (n=78 excluded)
- Large structural variants (>100 bp)
- Variants in non-coding regions >10 kb from gene body

Total downloaded: **431 variants**; after exclusion of 78 complex indels: **Final dataset: n=353 variants**

Variant Categorization

Variants were annotated using VEP v113 REST API with the following categories and counts:

Category	n
missense	125
frameshift	99
nonsense	40
splice_donor	22
promoter	15
3_prime_UTR	13
other	12

splice_region	9
intronic	9
5_prime_UTR	3
splice_acceptor	3
synonymous	3
Total	**353**

Category definitions:

- `splice_donor`, `splice_acceptor`: ±2 bp from exon boundary
- `splice_region`: ±8 bp from exon boundary (excluding donor/acceptor)
- `missense`, `nonsense`, `frameshift`
- `promoter`: -2000 to +200 bp from TSS
- `5_prime_UTR`, `3_prime_UTR`, `intronic`

Statistical Analysis

Clustering Analysis

SSIM clustering was assessed using:

- **Standard deviation:** $\sigma = \text{sqrt}(\Sigma(x_i - \mu)^2 / (n-1))$
- **Coefficient of variation:** $CV = \sigma / \mu$
- **Z-score normalization:** $z_i = (x_i - \mu) / \sigma$

Statistical significance of SSIM clustering for "Loop That Stayed" variants was evaluated via permutation test:

1. Randomly sample 3 variants from the full 353-variant dataset
2. Calculate SD of SSIM scores
3. Repeat 10,000 times
4. Empirical p-value: fraction of permutations with SD ≤ observed SD (0.0022)

Result: $p < 0.0001$ (none of 10,000 permutations achieved SD ≤ 0.0022)

ACMG Criteria Application

Variants were evaluated using ACMG/AMP 2015 guidelines (Richards et al.) with the following evidence:

PS3_moderate (Functional studies):

- ARCHCODE SSIM-based prediction (analytical mean-field model)
- Supporting evidence: qualitative consistency with published cohesin dynamics; no formal R^2 validation against experimental data is available
- Moderate strength (not strong) due to computational vs experimental nature

PM2 (Rarity):

- gnomAD v4.0 allele frequency
- Threshold: MAF < 0.0001 in all populations

PP3 (Multiple computational predictors):

- Conservation: PhyloP (vertebrate), PhastCons (mammalian)
- Structural: ARCHCODE SSIM
- Sequence: CADD, REVEL (supporting, though less weight for splice_region)

Point assignment:

- PS3_moderate: 4 points
- PM2: 2 points
- PP3: 1 point
- **Total: 7 points** (threshold for Likely Pathogenic: 6 points per ACMG)

ROC Analysis and Benign Variant Evaluation

To assess discriminative performance, the pathogenic dataset (353 variants) was supplemented with 750 Benign/Likely Benign HBB variants from ClinVar (queried via NCBI E-utilities API with significance filter "benign" OR "likely benign"). **All 1,103 variants were processed through a single unified TypeScript simulation engine** (`generate-unified-atlas.ts`), ensuring that `getEffectStrength()` receives only the variant's functional category and never the ClinVar classification label. This eliminates a pipeline discrepancy present in the initial analysis where pathogenic and benign variants were processed through different engines with incompatible perturbation scales. The combined cohort (n=1,103) was used for ROC analysis with ClinVar classification as ground truth (Pathogenic/LP = positive, Benign/LB = negative) and (1 − SSIM) as the continuous predictor. AUC, sensitivity, specificity, and Youden's J index were computed using scikit-learn v1.6. Optimal SSIM threshold was determined by maximizing J = Sensitivity + Specificity − 1.

Software and Code Availability

- **ARCHCODE simulator:** https://github.com/sergeeey/ARCHCODE (v1.1.0)
- **Analysis scripts:** TypeScript (Node.js v20), Python 3.11
- **Visualization:** matplotlib 3.8.2
- **Statistical analysis:** NumPy 1.26.3, SciPy 1.12.0
- **Random number generation:** SeededRandom class (Mersenne Twister, seed=2026)

All simulations were run on:

- **CPU:** AMD Ryzen 9 5900X (12 cores)
- **RAM:** 64 GB DDR4-3200
- **OS:** Windows 11 Pro (WSL2 for Python scripts)
- **Compute time:** < 1 second per variant (analytical approach), ~50 seconds total for 1,103 variants (unified pipeline)

Data Availability

All data supporting the findings of this study are available from the corresponding author upon reasonable request. Key datasets include:

- Unified variant analysis (353 pathogenic + 750 benign = 1,103 variants): `HBB_Unified_Atlas.csv`
- Legacy combined analysis (v1.0, dual-pipeline): `HBB_Combined_Atlas.csv`
- Pathogenic-only analysis (353 variants): `HBB_Clinical_Atlas_REAL.csv`
- Pearl analysis summary: `REAL_ATLAS_SUMMARY.json`
- VEP sequence-level predictions: `hbb_vep_results.csv` , `hbb_benign_vep_results.csv`
- ROC analysis (unified): `roc_unified.json`
- Source code: GitHub repository (see Software and Code Availability)

Methods section prepared for bioRxiv submission Last updated: 2026-02-28

Results

ClinVar HBB variant dataset

We downloaded 431 HBB variant records from ClinVar via NCBI E-utilities (esearch + esummary API, accessed 2026-02-28). After filtering for records that contained both reference and alternate allele information, 353 variants were retained for analysis (78 records were excluded due to missing allele data in the esummary response). The retained set comprised 12 molecular consequence categories: nonsense (40), frameshift (99), missense (125), splice_donor (22), splice_acceptor (3),

splice_region (9), promoter (15), 5_prime_UTR (3), 3_prime_UTR (13), intronic (9), synonymous (3), and other (12). ClinVar clinical significance labels in the retained set included Pathogenic (234), Pathogenic/Likely pathogenic (44), Likely pathogenic (46), Pathogenic with other interpretations (28), and Pathogenic/Likely pathogenic with other interpretations (1). No VUS remained in the final dataset after filtering. All downstream analyses used clinical significance as recorded in ClinVar without modification.

ARCHCODE simulation of 353 HBB variants

Each variant was simulated using the ARCHCODE analytical mean-field contact model with Kramer kinetics (parameters: $\alpha=0.92$, $\gamma=0.80$, k_base=0.002; see Methods for parameter provenance). Simulations covered a 30 kb window centered on the HBB locus (chr11:5,210,000–5,240,000, GRCh38) at 600 bp resolution (50 bins), incorporating 6 CTCF anchor sites and 5 annotated enhancer/regulatory elements (HBB proximal promoter, 3'HS1, and LCR-proximal elements). For each variant, a wild-type (WT) contact matrix and a mutant contact matrix were computed analytically. Structural disruption was quantified using the Structural Similarity Index (SSIM) between WT and mutant matrices, where SSIM = 1 indicates no structural change and SSIM approaching 0 indicates complete disruption of the contact pattern. A variant was classified as ARCHCODE-pathogenic if SSIM fell below the threshold of 0.95 (see Methods for threshold selection rationale).

ARCHCODE SSIM correlates with expected functional severity

Mean SSIM values showed a clear monotonic relationship with expected functional impact when variants were grouped by molecular consequence category (Table 1). Synonymous variants, which do not alter the protein sequence and are not expected to disrupt chromatin topology, returned the highest mean SSIM (0.9989, n=3). Intronic and 3'UTR variants likewise showed near-complete structural preservation (mean SSIM 0.9957 and 0.9942, respectively). Missense variants showed intermediate structural preservation (mean SSIM 0.9526, n=125), consistent with single amino-acid substitutions that do not typically remodel loop-anchor sequences. Splice-donor variants showed more substantial structural disruption (mean SSIM 0.9087, n=22), followed by frameshift (0.8919, n=99) and nonsense variants (0.8753, n=40). The complete rank order across all 12 categories is:

synonymous (0.9989) > intronic (0.9957) > 3'UTR (0.9942) > 5'UTR (0.9801) > other (0.9676) > splice_region (0.9641) > missense (0.9526) > promoter (0.9285) > splice_donor (0.9087) > splice_acceptor (0.9019) > frameshift (0.8919) > nonsense (0.8753)

This ordering is biologically expected: loss-of-function variant classes (nonsense, frameshift) produce the most severe chromatin disruption in the model, while conservative substitutions (synonymous, deep intronic) produce the least. The mean SSIM across all 353 variants was 0.9267 (per-variant values are available in Supplementary Table S1).

Table 1. ARCHCODE SSIM statistics by variant category.

Category	n	Mean SSIM	ARCHCODE Pathogenic (SSIM < 0.95)	%
nonsense	40	0.8753	40	100%
frameshift	99	0.8919	99	100%
splice_acceptor	3	0.9019	3	100%
splice_donor	22	0.9087	19	86%
promoter	15	0.9285	0	0%
missense	125	0.9526	0	0%
splice_region	9	0.9641	0	0%
other	12	0.9676	0	0%
5_prime_UTR	3	0.9801	0	0%

3_prime_UTR	13	0.9942	0	0%
intronic	9	0.9957	0	0%
synonymous	3	0.9989	0	0%
Total	**353**	**0.9267**	**161**	**45.6%**

Concordance and discordance with VEP sequence-based predictions

Across all 353 variants, ARCHCODE classified 161 (45.6%) as pathogenic and the sequence- based VEP pipeline classified 287 (81.3%) as pathogenic (mean VEP score 0.754 across the full dataset). Overall discordance — defined as variants where ARCHCODE and VEP verdicts differ — was observed in 130 variants (36.8% of the dataset). The discordance was strongly asymmetric: 128 variants were called pathogenic by VEP but not by ARCHCODE (VEP-only), and only 2 variants were called pathogenic by ARCHCODE but not by VEP (ARCHCODE-only).

This asymmetry is mechanistically expected. VEP integrates SIFT, consequence annotations (stop_gained, frameshift_variant, splice_donor_variant, etc.), and conservation scores — tools calibrated primarily to detect local sequence-level damage to proteins and splice signals. Most pathogenic HBB variants act through exactly these local mechanisms: amino- acid substitution (missense, n=125) or disruption of protein-coding reading frame (nonsense, frameshift, n=139 combined). ARCHCODE models 3D chromatin contact rearrangements within the 30 kb simulation window, which is most sensitive to changes at CTCF anchor sequences and enhancer elements — changes that predominantly occur in loss-of-function variants. Consequently, the 128 VEP-only discordant variants are enriched for missense and splice_region classes where the pathogenic mechanism operates at the protein or spliceosome level rather than through chromatin loop topology.

The 2 ARCHCODE-only variants (detected by ARCHCODE, missed by VEP) represent candidates for further investigation. Their genomic positions and variant details are provided in Supplementary Table S1; experimental validation would be required before clinical conclusions could be drawn.

Pearl variant discovery: structural disruption in VEP-blind loci

We defined a "pearl variant" as a variant satisfying two criteria simultaneously: VEP score < 0.30 (indicating VEP classifies it as low-impact or benign) AND SSIM < 0.95 (indicating ARCHCODE detects structural disruption). Twenty variants met both criteria (Table 2). These represent cases where sequence-based prediction is blind but the structural model detects disruption.

The 20 pearl variants fall into three groups by molecular context:

Group 1 — Promoter variants (15 of 20 pearls). Fifteen variants map to positions 5,227,099–5,227,172 on chr11, within the HBB proximal promoter region (SSIM range 0.927–0.930; VEP score 0.20 for all). VEP annotates these as *upstream_gene_variant*, a consequence term associated with low predicted impact in standard VEP weighting schemes. However, the ARCHCODE simulation places this region within a simulated enhancer element (LCR–HBB contact domain), and substitutions here reduce modeled enhancer–promoter contact scores, yielding SSIM values that fall below the 0.95 threshold. This represents the key complementarity between the two approaches: VEP does not model enhancer–promoter loop contacts and therefore cannot detect disruption of these interactions, whereas ARCHCODE's physics-based loop extrusion model is sensitive to changes at simulated anchor and enhancer sites. Whether this simulated sensitivity corresponds to genuine experimental disruption of chromatin contacts at these positions has not been validated and remains to be tested.

Group 2 — Missense variants at position 5,226,613 (3 of 20 pearls). Three variants at chr11:5,226,613 returned SSIM=0.949 and VEP=0.20. VEP annotates these as *coding_sequence_variant*, a term typically applied to complex indels where consequence prediction is ambiguous (poor annotation coverage). SSIM falls marginally below the 0.95 threshold. These are low-confidence pearls: the SSIM deviation is small (0.001 below threshold) and VEP annotation quality is reduced for complex indels.

Group 3 — Loss-of-function variants with low VEP scores (2 of 20 pearls). One frameshift variant at position 5,226,971 (SSIM=0.891, VEP=0.15) and one splice_acceptor variant at position 5,226,796 (SSIM=0.900, VEP=0.20) completed the pearl set. Their low VEP scores likely reflect incomplete VEP annotation rather than genuinely low pathogenicity; ClinVar records

these as Pathogenic. These variants are correctly identified by ARCHCODE (SSIM well below 0.95) but are included here for completeness as they technically satisfy the pearl criteria.

Representative ClinVar accessions in the pearl set include (but are not limited to): VCV002664746, VCV000811500, VCV000015208, VCV002024192, VCV000869358, VCV000015471, VCV000015470, VCV000869288, VCV000869290, VCV000015466. Full accession lists for all 20 pearl variants are provided in Supplementary Table S1.

Table 2. Top 5 pearl variants by ARCHCODE structural disruption (of 20 total; VEP < 0.30 AND SSIM < 0.95).

ClinVar_ID	HGVS_c	Category	ClinVar_Significance	SSIM	VEP	VEP_Consequen
VCV000869358	c.50dup	frameshift	Pathogenic	0.8915	0.15	synonymous_varian
VCV002024192	c.93-33_96delins...	splice_acceptor	Likely pathogenic	0.9004	0.20	coding_sequence_va
VCV000015471	c.-78A>G	promoter	Pathogenic/LP	0.9276	0.20	5_prime_UTR_varian
VCV000015470	c.-78A>C	promoter	Pathogenic	0.9276	0.20	5_prime_UTR_varian
VCV000036284	c.-136C>T	promoter	Pathogenic/LP	0.9277	0.20	5_prime_UTR_varian

Full list of 20 pearls sorted by SSIM: Supplementary Table S1 (manuscript/TABLE_S1_PEARLS.md).

Pearl summary by group:

Group	n	Positions (chr11)	Mean SSIM	Mean VEP	Molecular context
Promoter	15	5,227,099–5,227,172	0.928	0.20	5_prime_UTR_variant (HBB promoter)
Missense at 5226613	3	5,226,613	0.949	0.20	coding_sequence_variant (complex indel)
LoF (frameshift + splice)	2	5,226,796 / 5,226,971	0.896	0.18	frameshift / splice_acceptor
Total	**20**	—	**0.928**	**0.20**	—

ARCHCODE does not detect missense pathogenicity: an expected limitation

ARCHCODE classified 0% of missense variants as pathogenic (Table 1; n=125, mean SSIM 0.9526). This is the most prominent limitation of the current approach and must be stated explicitly. Missense variants are the largest category in the HBB ClinVar dataset and include well-validated pathogenic variants such as HbS (rs334, p.Glu7Val, responsible for sickle-cell disease). These variants act through protein structural perturbation — a mechanism entirely outside ARCHCODE's scope. ARCHCODE models chromatin contact topology, not protein folding or function. A single-nucleotide substitution in a coding exon does

not substantially displace CTCF anchor sequences or enhancer elements at 600 bp resolution within a 30 kb window, and therefore produces SSIM near 1.0 regardless of protein-level impact.

Similarly, ARCHCODE classified 0% of intronic (n=9), synonymous (n=3), 3'UTR (n=13), and splice_region (n=9) variants as pathogenic. While some of these (particularly splice_region and intronic) may carry genuine pathogenicity through mechanisms such as cryptic splice-site activation or branch-point disruption, the current ARCHCODE model does not include sequence- based splice scoring. These variants are therefore invisible to ARCHCODE unless they happen to coincide with a simulated CTCF anchor or enhancer element at 600 bp resolution.

These results reinforce the interpretation that ARCHCODE and sequence-based predictors such as VEP are orthogonal tools, each capturing a distinct mechanism. ARCHCODE is sensitive to structural disruption of chromatin loop domains (strongest signal for nonsense and frameshift at CTCF-adjacent positions) and, in the pearl analysis, potentially to enhancer–promoter contact disruption in the HBB promoter region. VEP is sensitive to protein-level changes, canonical splice-site disruption, and sequence conservation. The clinical utility of combining both approaches lies precisely in the complementarity between these two mechanistic blind spots.

ROC Analysis: Discrimination of Pathogenic vs Benign Variants

To evaluate ARCHCODE's discriminative performance, we expanded the dataset to include 750 Benign/Likely Benign HBB variants from ClinVar (predominantly intronic, n=658; synonymous, n=83), yielding a combined cohort of 1,103 variants (353 Pathogenic/LP + 750 Benign/LB). **All 1,103 variants were processed through a single unified TypeScript simulation engine**, ensuring label-blind physics: `getEffectStrength()` receives only the functional category, never the ClinVar classification. Using ClinVar classification as ground truth and (1 − SSIM) as the predictor, ROC analysis yielded AUC = 0.977 (Pathogenic mean SSIM = 0.927; Benign mean SSIM = 0.996; range 0.968–0.999).

Threshold evaluation by Youden's J index:

SSIM threshold	Sensitivity	Specificity	Youden J
< 0.994 (Youden optimum)	0.966	0.988	0.954
< 0.99	0.935	0.989	0.924
< 0.96	0.875	1.000	0.875
< 0.95 (current pearl threshold)	0.620	1.000	0.620
< 0.92 (LIKELY_PATHOGENIC)	0.456	1.000	0.456
< 0.85 (PATHOGENIC)	0.009	1.000	0.009

At stringent thresholds (SSIM < 0.96 and below), specificity remains 1.000. At the Youden optimum (SSIM < 0.994), 8 Benign variants fall below the threshold (all from the "other" and "5_prime_UTR" categories), yielding specificity = 0.988.

Note on pipeline unification: An initial analysis using separate processing pipelines for pathogenic (TypeScript) and benign (Python) variants yielded AUC = 1.000 due to a 10-fold difference in perturbation scaling between engines (e.g., intronic: 20% vs 2% occupancy reduction). After unifying all variants through a single engine, benign intronic SSIM decreased from 1.000 to 0.996 — matching pathogenic intronic SSIM (0.996) — and AUC dropped to 0.977. This confirms that the previous perfect AUC was a pipeline artifact. The current AUC of 0.977 reflects the genuine category-distribution difference between Pathogenic (enriched in nonsense/frameshift/splice) and Benign (enriched in intronic/ synonymous) cohorts, not independent variant-level prediction.

Critical caveat on AUC interpretation: The AUC reflects category-distribution differences between the Pathogenic and Benign cohorts (see Methods: Variant Introduction). Benign ClinVar variants are predominantly intronic and synonymous, which receive minimal occupancy perturbation by design. Pathogenic variants include nonsense, frameshift, and splice categories, which receive stronger perturbation. Within any single category, Pathogenic and Benign variants receive identical treatment — confirmed by within-category SSIM differences < 0.001 (e.g., intronic: Δ = 0.0002). The model's scientific

contribution lies in position-dependent structural discrimination within categories (e.g., which promoter variants disrupt enhancer–promoter contacts), not in category-level classification.

Quadrant analysis (SSIM threshold 0.95 / VEP threshold 0.30):

Quadrant	Description	Pathogenic	Benign	Total
Q1	Both detect (SSIM < 0.95, VEP ≥ 0.30)	199	0	199
Q2	ARCHCODE only / pearls (SSIM < 0.95, VEP < 0.30)	20	0	20
Q3	VEP only (SSIM ≥ 0.95, VEP ≥ 0.30)	95	41	136
Q4	Neither (SSIM ≥ 0.95, VEP < 0.30)	39	709	748

Critically, Q2 (ARCHCODE-only detections / pearl variants) contains 20 Pathogenic variants and 0 Benign variants under the unified pipeline. This demonstrates that pearl identification has zero false-positive rate among confirmed benign variants in the current dataset.

Summary

ARCHCODE simulation of 353 real ClinVar HBB variants demonstrates: (1) mean SSIM values rank variant categories in the biologically expected order from nonsense (most disruptive) to synonymous (least disruptive); (2) overall discordance with VEP is 36.8%, predominantly VEP-only (128 variants), reflecting VEP's sensitivity to protein-level mechanisms that lie outside ARCHCODE's scope; (3) 20 pearl variants with VEP score < 0.30 and SSIM < 0.95 suggest candidate cases for structural-level re-evaluation, with the strongest biological signal in promoter-region variants where the LCR–HBB enhancer–promoter contact model detects disruption not captured by VEP consequence annotation; (4) ROC analysis on a combined cohort of 1,103 variants (353 Pathogenic + 750 Benign), processed through a unified pipeline, yielded AUC = 0.977 with near-zero false positives among benign variants at stringent thresholds, reflecting category-distribution differences rather than independent sequence-based prediction; (5) ARCHCODE shows zero sensitivity to missense variants, its most important limitation for clinical variant classification.

All reported SSIM values and VEP scores are derived from computational models and have not been validated against experimental Hi-C data, RNA-seq, or patient phenotype data for the HBB locus. Experimental validation is required before any variant reclassification should be considered.

Results section — based on real ClinVar data (353 variants, downloaded via NCBI E-utilities) Word count: ~1,450 Last updated: 2026-02-28

Discussion

What ARCHCODE Reveals — and What It Cannot

We began this study with a clear question: can an analytical mean-field loop extrusion simulator identify pathogenic signal in HBB variants that sequence-based predictors miss? The answer is yes, but with important quantitative caveats.

ARCHCODE correctly stratifies 353 ClinVar HBB variants by functional severity: mean SSIM follows the biologically expected ranking from nonsense (0.8753) through missense (0.9526) to synonymous (0.9989). The 20 pearl variants — particularly the 15 promoter-region candidates — represent a concrete, testable prediction: that variants in the HBB proximal promoter region (chr11:5,227,099–5,227,172) disrupt modeled LCR–HBB enhancer–promoter contacts in ways invisible to VEP's consequence-based scoring.

However, the Hi-C validation result (r = 0.16, p = 0.301, not significant; n = 12 loci) is an essential caveat. ARCHCODE's SSIM scores reflect a theoretical model of cohesin dynamics, not measurements of actual chromatin contact frequencies. The weak Hi-C correlation indicates that the current parameter set does not quantitatively reproduce experimental contact maps at the HBB locus. This does not invalidate the comparative analysis — the ordering of variant categories is internally consistent —

but it does mean that SSIM values should be interpreted as relative disruption scores within the model, not as absolute predictions of chromatin structure change.

We have not "confirmed" the "Loop That Stayed" mechanism. We have generated a computational prediction that specific promoter-region HBB variants show structural disruption by our model. Whether this reflects genuine regulatory biology requires experimental testing.

The Structural Blind Spot of Sequence-Based Prediction

The 130 discordant variants (36.8% of dataset) illuminate the complementary nature of structural and sequence-based approaches. The 128 VEP-only variants — those called pathogenic by VEP but not by ARCHCODE — consist predominantly of missense variants acting through protein structural perturbation, a mechanism ARCHCODE cannot detect. This is expected and appropriate: ARCHCODE does not model protein folding.

The reverse discordance is more interesting: 2 variants and the 20 pearl candidates were detected by ARCHCODE but assigned low pathogenicity by VEP. For the promoter-region pearls, the mechanism is clear in model terms: VEP annotates upstream_gene_variant as low-impact because the standard consequence weighting does not account for LCR–HBB contact disruption. ARCHCODE's physics-based model, by contrast, is sensitive to changes at positions it has annotated as enhancer elements, regardless of whether these positions match a recognized splice or coding motif.

This complementarity mirrors the relationship between different imaging modalities in clinical medicine. VEP is sensitive to sequence-level defects in protein-coding and canonical splice sequences. ARCHCODE is sensitive to regulatory topology — changes at simulated CTCF anchors and enhancer elements within the 30 kb simulation window. A variant outside both detection ranges will be missed by both; a variant detected by only one provides hypothesis-generating signal for follow-up.

Model independence: ARCHCODE is mechanistically orthogonal to VEP. ARCHCODE receives no input from VEP scores, SIFT predictions, ClinVar classifications, or any sequence-based tool. The two models operate on fundamentally different biological layers: VEP evaluates protein- coding impact and splice motif disruption from primary sequence; ARCHCODE simulates cohesin- mediated loop extrusion and chromatin contact topology from occupancy profiles. Their outputs are statistically compared post hoc but are generated independently. The absence of missense sensitivity in ARCHCODE is therefore a topological feature of the model, not a failure — missense variants that cause disease through protein misfolding, loss of catalytic activity, or dominant-negative mechanisms operate entirely at the protein level, a dimension invisible to chromatin topology simulation. For missense variants, sequence-based VEP tools remain the appropriate primary tool. ARCHCODE provides orthogonal evidence *specifically* for regulatory, promoter, and structural variants where 3D genome organization mediates the mechanism.

"The Loop That Stayed" as Theoretical Framework

The "Loop That Stayed" framework posits that stable cohesin-mediated loops can confine splice regulatory defects by limiting spliceosome access to trans-acting factors outside the loop domain. Our computational results are consistent with this framework in a narrow sense: splice_region and promoter variants within the simulated LCR–HBB loop domain show structural disruption signals (SSIM < 0.95) that VEP does not detect.

However, we emphasize that this is a theoretical mechanism, not an experimentally confirmed pathway. The simulation shows disruption of a modeled contact pattern; it does not demonstrate that any spliceosome is topologically confined. The following chain of inference is required to connect our computational results to a clinical finding:

1. ARCHCODE simulation predicts structural disruption at promoter-region positions (SSIM ~0.928)
2. Experimental Capture Hi-C would need to confirm reduced enhancer–promoter contact frequency
3. RT-PCR in erythroid cells (K562, HUDEP-2) would need to confirm aberrant splicing or reduced transcription
4. Patient genotype-phenotype correlation would need to link the variant to disease severity

We have completed step 1. Steps 2–4 are required before any variant reclassification.

Limitations

1. Computational model only; Hi-C validation weak (r = 0.16, p = ns). ARCHCODE simulations remain *in silico* predictions. The weak Hi-C correlation is the most important quantitative limitation: our model does not yet reproduce experimental

contact frequencies with statistical reliability.

2. Parameters manually calibrated, not fitted. $\alpha=0.92$ and $\gamma=0.80$ are estimated from literature ranges, not fitted to HBB-specific FRAP data. Cell-type-specific differences between GM12878 (our ChIP-seq source) and erythroid cells (the relevant clinical context) may substantially alter loop dynamics.

3. Analytical mean-field model. The model lacks polymer physics baseline, A/B compartmentalization, non-CTCF enhancer loops (YY1, LDB1), and cell-to-cell heterogeneity. These omissions likely explain the weak Hi-C correlation.

4. VEP instead of SpliceAI. The SpliceAI API was unreachable during the study period. SpliceAI provides higher-resolution splice disruption prediction than VEP/SIFT and would likely improve the sequence-based baseline, potentially reclassifying some current pearls.

5. No patient-level validation. No patients with these variants and phenotype data are available in ClinVar records to validate computational predictions against clinical reality.

6. Pearl variants not experimentally validated. The 20 pearl variants are computational candidates. None have been tested by RT-PCR, Capture Hi-C, or functional assay in this study.

Path to Clinical Translation

Clinical reclassification of any variant based on ARCHCODE predictions alone would be premature. The appropriate translational pathway is:

Step 1 (immediate): Release pearl variant list and ARCHCODE analysis as bioRxiv preprint. Invite experimental groups with HBB expertise and erythroid cell models to test predictions.

Step 2 (3–6 months): RT-PCR in K562 or HUDEP-2 cells for the 15 promoter-region pearl variants. This directly tests whether VEP-blind promoter variants cause measurable changes in HBB transcript abundance or splicing.

Step 3 (6–12 months): Capture Hi-C at the HBB locus in erythroid cells, comparing wild-type to variants with the largest SSIM deviations. This would validate or refute the model's contact disruption predictions and improve the $r = 0.16$ correlation baseline.

Step 4 (if experimental evidence positive): Apply ACMG PS3 criteria (functional studies) with experimental RT-PCR data and submit evidence to ClinVar for affected accessions.

We specifically do not recommend clinical reclassification based on the computational results reported here.

Broader Implications

Despite the quantitative limitations, the ARCHCODE approach demonstrates a proof-of-concept for orthogonal structural scoring. The identification of 20 pearl variants provides a concrete, size-limited prioritization list for experimental groups — far more tractable than screening all 353 variants individually.

For the broader field of variant interpretation, this work illustrates that different computational tools capture different biological dimensions. Adding a structural simulation layer to existing sequence-based pipelines does not require high Hi-C correlation accuracy to provide hypothesis-generating utility: even a model with weak absolute accuracy can correctly rank variants by structural disruption severity and flag specific candidates where sequence-based methods have known blind spots.

The most important recommendation we can make is methodological: validation against experimental data should precede clinical deployment. The $r = 0.16$ result is not a publication failure — it is the honest establishment of a baseline that the next model iteration must exceed.

Discussion section prepared for bioRxiv submission Word count: ~1,100 words Last updated: 2026-02-28

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[Full bibliography in BibTeX available in repo.]

Supplementary Table S1: Comprehensive Analysis of 353 HBB Variants

Table Legend

Complete dataset of 353 β-globin (*HBB*) pathogenic and likely pathogenic variants analyzed using ARCHCODE (analytical mean-field 3D chromatin simulation) and Ensembl VEP v113 (sequence-based predictor). Variants are sorted by SSIM ascending (most structurally disruptive first).

Full dataset file: `HBB_Clinical_Atlas_REAL.csv` (353 rows, available from corresponding author or GitHub repository)

Columns in full dataset:

1. **ClinVar_ID**: ClinVar accession (VCV format)
2. **Position**: Genomic coordinate on chr11 (GRCh38/hg38)
3. **Ref / Alt**: Reference and alternate alleles
4. **Category**: Variant functional category (VEP annotation; see Methods for definitions)
5. **ARCHCODE_SSIM**: Structural Similarity Index (WT vs Mutant contact matrices)
   - Range: [0,1], higher = more similar to WT (less structural disruption)
   - Thresholds: SSIM < 0.85 PATHOGENIC; 0.85–0.92 LIKELY_PATHOGENIC; 0.92–0.96 VUS; 0.96–0.99 LIKELY_BENIGN; ≥ 0.99 BENIGN
6. **VEP_Score**: Ensembl VEP v113 pathogenicity score (consequence + SIFT integration)
   - Range: [0,1], higher = more pathogenic

- Thresholds: ≥ 0.50 Pathogenic/Likely Pathogenic; < 0.50 Benign/Likely Benign/VUS

7. **ARCHCODE_Verdict**: ARCHCODE classification based on SSIM threshold
8. **VEP_Verdict**: VEP classification based on score threshold
9. **Discordant**: YES if verdicts differ between methods, NO otherwise
10. **Pearl**: YES if VEP score < 0.30 AND SSIM < 0.95 (both criteria required)

Summary Statistics by Category

Category	n	Mean SSIM	Mean VEP Score	% ARCHCODE Pathogenic
nonsense	40	0.8753	0.90	100%
frameshift	99	0.8919	0.90	100%
splice_acceptor	3	0.9019	0.95	100%
splice_donor	22	0.9087	0.95	86%
promoter	15	0.9285	0.20	0%
missense	125	0.9526	~0.70	0%
splice_region	9	0.9641	0.50	0%
other	12	0.9676	varies	0%
5_prime_UTR	3	0.9801	0.20	0%
3_prime_UTR	13	0.9942	0.15	0%
intronic	9	0.9957	0.10	0%
synonymous	3	0.9989	0.05	0%
Total	**353**	**0.9267**	**0.754**	**45.6%**

Interpretation: ARCHCODE SSIM and VEP are anti-correlated by design for non-coding categories: promoter, UTR, and intronic variants have low VEP scores (not annotated as high-impact by sequence analysis) but show moderate SSIM disruption in the structural model (0.92–0.98 range). This is the source of the pearl candidates. Concordance is highest for loss-of-function classes (nonsense, frameshift) where both methods detect severe disruption.

Discordance Summary

Discordance type	n	% of total	Dominant variant classes
VEP-only pathogenic	128	36.3%	missense (125), splice_region (3)
ARCHCODE-only pathogenic	2	0.6%	frameshift (1), splice_acceptor (1)
Concordant pathogenic	159	45.0%	nonsense, frameshift, splice donor/acceptor
Concordant benign/VUS	64	18.1%	synonymous, intronic, UTR, promoter
Total	**353**	**100%**	

Pearl Variants: Real ClinVar Accessions

The following variants met both pearl criteria (VEP score < 0.30 AND SSIM < 0.95). ClinVar accession numbers are real; they can be verified at: https://www.ncbi.nlm.nih.gov/clinvar/

Group 1: Promoter-Region Pearls (n=15)

These variants map to the HBB proximal promoter region (chr11:5,227,099–5,227,172, GRCh38). VEP annotates them as `upstream_gene_variant` (low predicted impact). ARCHCODE places this region within a simulated LCR–HBB enhancer–promoter contact domain and detects disruption (mean SSIM 0.928, mean VEP 0.20).

ClinVar_ID	Position (chr11)	Category	SSIM	VEP Score	ClinVar Significance
VCV000015471	5,227,099	promoter	0.928	0.20	Pathogenic/Likely path.
VCV000015470	5,227,099	promoter	0.928	0.20	Pathogenic
VCV000869288	5,227,100	promoter	0.929	0.20	Pathogenic
VCV000869290	5,227,101	promoter	0.928	0.20	Pathogenic
VCV000015466	5,227,102	promoter	0.929	0.20	Pathogenic
VCV000801184	5,227,142	promoter	0.928	0.20	Pathogenic/Likely path.
VCV000036284	5,227,157	promoter	0.928	0.20	Pathogenic/Likely path.

7 of 15 promoter pearl accessions shown. All 15 with exact positions and alleles are available in HBB_Clinical_Atlas_REAL.csv.

Group 2: Missense Pearls at chr11:5,226,613 (n=3)

Three variants at position 5,226,613 are annotated as `coding_sequence_variant` by VEP (incomplete consequence prediction for complex indels). SSIM = 0.949 (marginally below the 0.95 threshold). These are low-confidence pearls.

ClinVar_ID	Position (chr11)	Category	SSIM	VEP Score	ClinVar Significance
VCV002664746	5,226,613	missense	0.949	0.20	VUS
(2 additional accessions in HBB_Clinical_Atlas_REAL.csv)					

Group 3: Loss-of-Function Pearls with Low VEP Annotation (n=2)

These LoF variants have low VEP scores due to incomplete annotation, not because they are genuinely low-impact. ClinVar records both as Pathogenic.

ClinVar_ID	Position (chr11)	Category	SSIM	VEP Score	ClinVar Significance
VCV002024192	5,226,796	splice_acceptor	0.900	0.20	Pathogenic
VCV000869358	5,226,971	frameshift	0.891	0.15	Pathogenic

ACMG Evidence Assessment for Pearl Variants (Exploratory)

Important disclaimer: The following ACMG assessment is exploratory and computational only. No experimental functional data exists for these variants in this study. Clinical reclassification requires experimental evidence (PS3 from functional studies with actual RT-PCR/RNA data).

For promoter-region pearl variants (Group 1) where ClinVar records Pathogenic:

ACMG Criterion	Evidence	Strength
PS3_moderate	ARCHCODE SSIM-based structural disruption prediction	Moderate (computational only)
PM2	gnomAD v4.0 MAF < 0.0001 (consistent with rare disease variant)	Moderate
PP3	ARCHCODE structural prediction + PhyloP conservation	Supporting
Total	**PS3_mod (4) + PM2 (2) + PP3 (1) = 7 points**	**Meets LP threshold (≥6)**

This assessment supports the existing ClinVar Pathogenic/Likely Pathogenic classifications for Group 1 variants, but does not independently justify reclassification of VUS variants without experimental confirmation.

Data Availability

Unified dataset (v2.0): `HBB_Unified_Atlas.csv` (1,103 rows: 353 Pathogenic + 750 Benign, single pipeline)

Legacy combined dataset (v1.0): `HBB_Combined_Atlas.csv` (1,103 rows, dual-pipeline; preserved for reproducibility)

Pathogenic-only dataset: `HBB_Clinical_Atlas_REAL.csv` (353 rows × 10 columns)

Pearl analysis summary: `REAL_ATLAS_SUMMARY.json`

VEP predictions: `hbb_vep_results.csv` (pathogenic), `hbb_benign_vep_results.csv` (benign)

ROC analysis (unified): `roc_unified.json`

ROC analysis (v1.0 legacy): `roc_analysis.json`

Format: Comma-separated values (CSV), UTF-8 encoding

Repository: https://github.com/sergeeey/ARCHCODE

Contact: sergeikuch80@gmail.com

Methods Summary (Supplementary)

ARCHCODE simulation:

- Analytical mean-field contact computation (no Monte Carlo sampling)
- Kramer kinetics: $\alpha=0.92$, $\gamma=0.80$ (manually calibrated; not fitted to FRAP)
- 50 bins × 600 bp = 30 kb HBB locus window
- SSIM calculated on upper triangular matrix (k=1, excluding diagonal)
- Seed=2026 for reproducibility

VEP predictions:

- Ensembl VEP v113 REST API
- Consequence-based scoring + SIFT integration for missense
- SpliceAI not available (API unreachable during study period)

Statistical analysis:

- Permutation testing (10,000 iterations) for SSIM clustering significance
- ACMG/AMP 2015 guidelines for exploratory evidence assessment
- Discordance defined as ARCHCODE verdict ≠ VEP verdict (pathogenic vs benign/VUS)

Known limitations for interpretation:

- Hi-C validation: r = 0.16, p = 0.301 (not significant; n = 12 loci)
- Parameters manually calibrated, not fitted to HBB locus
- GM12878 ChIP-seq used; erythroid-specific data not available

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Supplementary Table S1 prepared for bioRxiv submission Last updated: 2026-02-28 Corresponding author: Sergey V. Boyko (sergeikuch80@gmail.com)